

Born approximation 2

We will now derive the Green's function that was used in the previous section.

Let $G(\mathbf{x})$ be a Green's function such that

$$(\nabla^2 + k^2) G(\mathbf{x}) = \delta(\mathbf{x}) \quad (1)$$

Let $g(\mathbf{y})$ be the Fourier transform of $G(\mathbf{x})$ such that

$$G(\mathbf{x}) = \frac{1}{(2\pi)^{3/2}} \int \exp(i\mathbf{x} \cdot \mathbf{y}) g(\mathbf{y}) d\mathbf{y} \quad (2)$$

By equations (1) and (2)

$$(\nabla^2 + k^2) \left[\frac{1}{(2\pi)^{3/2}} \int \exp(i\mathbf{x} \cdot \mathbf{y}) g(\mathbf{y}) d\mathbf{y} \right] = \delta(\mathbf{x})$$

By linearity

$$\frac{1}{(2\pi)^{3/2}} \int (\nabla^2 + k^2) \exp(i\mathbf{x} \cdot \mathbf{y}) g(\mathbf{y}) d\mathbf{y} = \delta(\mathbf{x})$$

Noting that ∇^2 operates on $\mathbf{x}$ and not $\mathbf{y}$ we have

$$\nabla^2 \exp(i\mathbf{x} \cdot \mathbf{y}) g(\mathbf{y}) = -|\mathbf{y}|^2 \exp(i\mathbf{x} \cdot \mathbf{y}) g(\mathbf{y})$$

Hence

$$\frac{1}{(2\pi)^{3/2}} \int (-|\mathbf{y}|^2 + k^2) \exp(i\mathbf{x} \cdot \mathbf{y}) g(\mathbf{y}) d\mathbf{y} = \delta(\mathbf{x})$$

By the identity

$$\delta(\mathbf{x}) = \frac{1}{(2\pi)^3} \int \exp(i\mathbf{x} \cdot \mathbf{y}) d\mathbf{y}$$

we have

$$\frac{1}{(2\pi)^{3/2}} \int (-|\mathbf{y}|^2 + k^2) \exp(i\mathbf{x} \cdot \mathbf{y}) g(\mathbf{y}) d\mathbf{y} = \frac{1}{(2\pi)^3} \int \exp(i\mathbf{x} \cdot \mathbf{y}) d\mathbf{y}$$

Hence

$$g(\mathbf{y}) = \frac{1}{(2\pi)^{3/2} (k^2 - |\mathbf{y}|^2)}$$

Substitute for $g(\mathbf{y})$ in (2) to obtain

$$G(\mathbf{x}) = \frac{1}{(2\pi)^3} \int \frac{\exp(i\mathbf{x} \cdot \mathbf{y})}{k^2 - |\mathbf{y}|^2} d\mathbf{y}$$

Change to polar coordinates where $x = |\mathbf{x}|$ and $r = |\mathbf{y}|$.

$$G(\mathbf{x}) = \frac{1}{(2\pi)^3} \int_0^\infty \int_0^\pi \int_0^{2\pi} \frac{\exp(ixr \cos \theta)}{k^2 - r^2} r^2 \sin \theta dr d\theta d\phi$$

Integrate over ϕ (multiply by 2π).

$$G(\mathbf{x}) = \frac{1}{(2\pi)^2} \int_0^\infty \int_0^\pi \frac{\exp(ixr \cos \theta)}{k^2 - r^2} r^2 \sin \theta \, dr \, d\theta$$

For the integral over θ use change of variable $u = \cos \theta$ and $du = -\sin \theta \, d\theta$ to obtain

$$G(\mathbf{x}) = \frac{1}{(2\pi)^2} \int_0^\infty \int_1^{-1} -\frac{\exp(ixru)}{k^2 - r^2} r^2 \, dr \, du$$

By the integral

$$\int_1^{-1} -\exp(ixru) \, du = \frac{2 \sin(xr)}{xr} \quad (3)$$

we have

$$G(\mathbf{x}) = \frac{1}{2\pi^2 x} \int_0^\infty \frac{\sin(xr)}{k^2 - r^2} r \, dr$$

Noting that $r \sin(xr)$ is an even function of r we can extend the lower limit as

$$G(\mathbf{x}) = \frac{1}{4\pi^2 x} \int_{-\infty}^\infty \frac{\sin(xr)}{k^2 - r^2} r \, dr$$

Negate the denominator.

$$G(\mathbf{x}) = \frac{1}{4\pi^2 x} \int_{-\infty}^\infty -\frac{\sin(xr)}{r^2 - k^2} r \, dr$$

Change the sine function to exponential form and factor the denominator.

$$G(\mathbf{x}) = \frac{i}{8\pi^2 x} \left[\int_{-\infty}^\infty \frac{\exp(ixr)}{(r+k)(r-k)} r \, dr - \int_{-\infty}^\infty \frac{\exp(-ixr)}{(r+k)(r-k)} r \, dr \right] \quad (4)$$

By Cauchy's integral formula we have

$$\int_{-\infty}^\infty \frac{r \exp(ixy)}{r+k} \frac{1}{r-k} \, dr = i\pi \exp(ikx)$$

and

$$\int_{-\infty}^\infty \frac{r \exp(-ixy)}{r-k} \frac{1}{r+k} \, dr = -i\pi \exp(ikx)$$

Hence

$$G(\mathbf{x}) = -\frac{\exp(ikx)}{4\pi x} \quad (5)$$

where

$$x = |\mathbf{x}|$$

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